An Elementary Proof of Sun's Weighted Prime-Square Congruence

A reciprocal binomial sum and a reflected WZ certificate

Research manuscript prepared with ChatGPT for Keio

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Abstract

Let $a_n = \sum_k (-4)^k \binom{n}{k}^2 \binom{n-k}{k}^2$. We prove that, for every odd prime p, $\sum_{n=0}^{p-1}(20n+17)a_n \equiv p(7 + 10(\frac{-1}{p})) \pmod{p^2}$, the second assertion of Sun's Conjecture 5.8(ii). A universal congruence for sums of squared binomial coefficients reduces the problem to a reciprocal Ramanujan-type finite sum. Although its individual summands can have a factor of p in the denominator, an explicit rational WZ certificate transforms the total into integral boundary terms. The two surviving endpoints are congruent to 3 and 4 modulo p; every other term vanishes at the required precision. The proof includes $p = 3$ uniformly and uses no other supercongruence or all-integer divisibility theorem as an input. We also deduce that p^2 divides the original weighted sum if and only if $p \in \{3, 17\}$. The established WZ antecedents are credited, and historical priority is not asserted.

1 The unchanged assertion

Define

$$a_n = \sum_{k=0}^{\lfloor n/2 \rfloor} (-4)^k \binom{n}{k}^2 \binom{n-k}{k}^2, \qquad S(N) = \sum_{n=0}^{N-1} (20n + 17)a_n. \tag{1}$$

The sequence is $W_n(1, -4)$ in Sun's notation and OEIS A179524 [1, 2]. The second assertion of Conjecture 5.8(ii) in [1, printed p. 33] predicts

$$S(p) \equiv p\left(7 + 10\left(\frac{-1}{p}\right)\right) \pmod{p^2} \qquad (p \text{ odd prime}). \tag{2}$$

Here $(\frac{-1}{p}) = (-1)^{(p-1)/2}$ is the Legendre symbol. The sum in (2) has exactly p terms.

The preceding research manuscript [6] treated the first assertion of Conjecture 5.8(ii), namely $N \mid S(N)$ for every positive integer N. The present proof of (2) is independent of that manuscript. Its ingredients are finite binomial identities, an explicit rational telescoping certificate, and elementary prime-adic valuation estimates. In particular, every ingredient needed for the new conclusion is proved below.

The certificate was found by reflecting familiar Ramanujan WZ kernels. Such kernels and the WZ method have established antecedents in Guillera's work [3], Sun's proof of a Ramanujan supercongruence [4, Section 3], and the parameter pair reproduced by Hou, He, and Wang [5, arXiv v4, (4.1)]. We give the resulting rational certificate directly, so its validity does not require analytic continuation or convergence of an infinite hypergeometric series.

2 The companion congruence modulo a prime square

In this section we prove the second assertion in Sun's Conjecture 5.8(ii). The proof uses an explicit rational telescoping certificate; it does not require the all-integer divisibility theorem or a separate Ramanujan supercongruence as an input.

Theorem 2.1. *For every odd prime p,*

$$\sum_{n=0}^{p-1}(20n+17)a_n \equiv p\left(7+10\left(\frac{-1}{p}\right)\right) \pmod{p^2}.$$

We use congruences between rational numbers in the localization $\mathbb{Z}_{(p)}$. Every displayed rational congruence below is justified by verifying the relevant denominators or by an explicit valuation. Write $b_m = \binom{2m}{m}$.

2.1 A universal inner-sum congruence

Lemma 2.2. *Let p be an odd prime and $0 \le m < p$. Then*

$$\sum_{n=m}^{p-1}\binom{n}{m}^2 \equiv \frac{p}{(2m+1)b_m} \pmod{p^2}.$$

The rational number on the right belongs to $\mathbb{Z}_{(p)}$, including the cases in which its displayed denominator is divisible by p.

Proof. The product linearization and the hockey-stick identity give

$$\binom{n}{m}^2 = \sum_{j=0}^{m}\binom{m}{j}\binom{m+j}{j}\binom{n}{m+j}, \qquad \sum_{n=m}^{p-1}\binom{n}{m}^2 = \sum_{j=0}^{m} d_j\binom{p}{m+j+1},$$

where $d_j = \binom{m}{j}\binom{m+j}{j}$. The product linearization counts ordered pairs of m-subsets by the size $m+j$ of their union. We compare the last sum termwise with

$$p\sum_{j=0}^{m}\frac{(-1)^{m+j}d_j}{m+j+1}.$$

Put $r = m+j+1$. If $r < p$, use $\binom{p}{r} \equiv (-1)^{r-1}p/r \pmod{p^2}$. If $r = p$, the compared terms are exactly equal, since p is odd. If $r > p$, the binomial coefficient on the left vanishes, whereas d_j is divisible by p: indeed $p \le m+j \le 2p-2$ and both $m, j < p$. Also $p < r < 2p$, so the compared term on the right is divisible by p^2.

For completeness, the required finite rational identity is

$$\sum_{j=0}^{m}\frac{(-1)^{m+j}}{m+j+1}\binom{m}{j}\binom{m+j}{j} = \frac{1}{(2m+1)b_m}.$$

To prove it, expand

$$\frac{1}{m!}\frac{d^m}{dx^m}\left(x^m(1-x)^m\right) = \sum_{j=0}^{m}(-1)^j\binom{m}{j}\binom{m+j}{j}x^j.$$

Multiply by $(-1)^m x^m$, integrate from zero to one, and integrate by parts m times. The boundary terms vanish, and the result is $\int_0^1 x^m(1-x)^m\,dx = m!^2/(2m+1)!$.

Finally, $v_p((2m+1)b_m) \le 1$. For $m < (p-1)/2$ it is zero; for $m = (p-1)/2$ the factor $2m+1 = p$ contributes one; and for $m > (p-1)/2$ the central binomial coefficient contributes one while $2m+1$ is a unit. Thus the claimed rational congruence is well defined. □

2.2 An exact rational telescoping certificate

For nonnegative integers n, k, define

$$c_n = \frac{(-4)^n b_n^2}{(4n+1)b_{2n}}, \qquad \mathcal{H}(n,k) = c_n \frac{(-2n-\frac{1}{2})_k (n+\frac{1}{2})_k}{(\frac{1}{2})_k^2},$$

where $(a)_k = a(a+1)\cdots(a+k-1)$ and $(a)_0 = 1$. Put

$$\mathcal{D}(n,k) = (20n+2k+7)\mathcal{H}(n,k), \qquad \mathcal{E}(n,k) = \frac{16(2n+1)^2}{4n-2k+3}\mathcal{H}(n,k),$$

and set $\mathcal{E}(-1,k) = 0$.

Lemma 2.3 (Rational certificate)**.** *For every $n \geq 0$ and every $k \geq 1$,*

$$\mathcal{D}(n,k) - \mathcal{D}(n,k-1) = \mathcal{E}(n,k) - \mathcal{E}(n-1,k).$$

Consequently, for every positive integer h,

$$\sum_{n=0}^{h} (20n+7)c_n = \sum_{n=0}^{h} \mathcal{D}(n,h) - \sum_{k=1}^{h} \mathcal{E}(h,k). \tag{3}$$

Proof. For $n \geq 1$, successive-term ratios give

$$\frac{\mathcal{H}(n,k-1)}{\mathcal{H}(n,k)} = \frac{(2k-1)^2}{(2k-4n-3)(2n+2k-1)},$$

$$\frac{\mathcal{H}(n-1,k)}{\mathcal{H}(n,k)} = -\frac{n(4n-2k-1)(4n-2k+1)}{8(2n-1)^2(2n+2k-1)}.$$

After division by $\mathcal{H}(n,k)$, the asserted certificate is the rational identity

$$20n+2k+7 - \frac{(20n+2k+5)(2k-1)^2}{(2k-4n-3)(2n+2k-1)} = \frac{16(2n+1)^2}{4n-2k+3} + \frac{2n(4n-2k+1)}{2n+2k-1},$$

which follows by clearing the two linear denominators. For $n = 0$, use $\mathcal{H}(0,0) = 1$ and $\mathcal{H}(0,k) = -1/(2k-1)$ for $k \geq 1$ to check the same identity directly with $\mathcal{E}(-1,k) = 0$. Summing over $0 \leq n \leq h$ and $1 \leq k \leq h$ gives (3). □

Theorem 2.4 (Reciprocal Ramanujan sum)**.** *For every odd prime p, with $h = (p-1)/2$,*

$$R_p := \sum_{n=0}^{h} \frac{(20n+7)(-4)^n b_n^2}{(4n+1)b_{2n}} \in \mathbb{Z}_{(p)}, \qquad R_p \equiv 7 \pmod{p}.$$

Proof. We evaluate the right-hand side of (3) modulo p. For $0 \leq n \leq h$,

$$v_p(c_n) = \begin{cases} 0, & 4n+1 < p, \\ -1, & 4n+1 \geq p. \end{cases}$$

Indeed b_n is a unit, and exactly one factor of p occurs in $(4n+1)b_{2n}$ in the second range. It comes from $4n+1$ when $4n+1 = p$, and from b_{2n} otherwise.

For $1 \leq n < h$, the product $(n+\frac{1}{2})_h$ contains exactly one factor divisible by p, namely $p/2$. If $4n+1 \geq p$, the other product $(-2n-\frac{1}{2})_h$ also contains exactly one such factor, namely $-p/2$. If $4n+1 < p$, it contains none. The denominator $(\frac{1}{2})_h^2$ is a unit. Therefore

$$v_p(\mathcal{H}(n,h)) = 1 \quad (1 \leq n < h), \qquad \mathcal{D}(n,h) \in p\mathbb{Z}_{(p)}.$$

This covers both ranges, including the possible equality $4n+1=p$.

For $1 \le k \le h$, c_h has valuation -1, $(h+\frac{1}{2})_k = (p/2)_k$ has valuation one, and $(-2h-\frac{1}{2})_k = (\frac{1}{2}-p)_k$ and $(\frac{1}{2})_k$ are units. Thus $\mathcal{H}(h,k)$ is a unit. Since

$$4h-2k+3 = 2p+1-2k$$

lies between $p+2$ and $2p-1$, it is a unit as well. The factor $(2h+1)^2 = p^2$ in $\mathcal{E}(h,k)$ now gives

$$\mathcal{E}(h,k) \in p^2\mathbb{Z}_{(p)}.$$

Only the two endpoint terms remain. At $n=0$,

$$\mathcal{D}(0,h) = -\frac{2h+7}{2h-1} = -\frac{p+6}{p-2} \equiv 3 \pmod{p}.$$

Cancellation of the finite products at $n=h$ gives the exact identity

$$\mathcal{H}(h,h) = 4^h b_h \prod_{i=1}^{h-1} \frac{p+2i}{2i+1}.$$

All denominators here are units. Also

$$\prod_{i=1}^{h-1} \frac{2i}{2i+1} = \frac{4^h}{(2h)b_h}.$$

Consequently, by Fermat's little theorem,

$$\mathcal{H}(h,h) \equiv \frac{16^h}{p-1} \equiv -1 \pmod{p},$$

and hence

$$\mathcal{D}(h,h) = (22h+7)\mathcal{H}(h,h) = (11p-4)\mathcal{H}(h,h) \equiv 4 \pmod{p}.$$

Every term on the right-hand side of (3) is now p-integral, and its reduction is $3+4=7$. The case $p=3$ is included: the interior range is empty and the displayed endpoint denominators are still units. □

2.3 Returning to the original weighted sum

Proof of Theorem 2.1. Put $\gamma_k = (-4)^k b_k^2$ and $h=(p-1)/2$. The elementary factorial identity $\binom{n}{2k}b_k = \binom{n}{k}\binom{n-k}{k}$ gives

$$a_n = \sum_k \gamma_k \binom{n}{2k}^2.$$

For $N > m \ge 0$, direct telescoping gives

$$\sum_{n=m}^{N-1} (2n-m+1)\binom{n}{m}^2 = \frac{N^2}{m+1}\binom{N-1}{m}^2. \tag{4}$$

Indeed, set $f_m(n) = (n-m)^2\binom{n}{m}^2/(m+1)$ for $n \ge m$. Then $f_m(n+1) = (n+1)^2\binom{n}{m}^2/(m+1)$, so the summand is $f_m(n+1)-f_m(n)$. The lower boundary $f_m(m)$ is zero, and the upper boundary has the displayed form. Using $20n+17 = 10(2n-2k+1)+(20k+7)$, interchange the finite sums and apply (4) to obtain

$$S(p) = 10p^2 \sum_{k=0}^{h} \frac{\gamma_k}{2k+1}\binom{p-1}{2k}^2 + \sum_{k=0}^{h}(20k+7)\gamma_k \sum_{n=2k}^{p-1}\binom{n}{2k}^2.$$

In the first sum, all terms with $k < h$ vanish modulo p^2. The remaining term is $10p\gamma_h$. Since

$$b_h = \binom{p-1}{h} \equiv (-1)^h \pmod{p}, \qquad (-4)^h \equiv (-1)^h \pmod{p},$$

we have $\gamma_h \equiv (-1)^h = (\frac{-1}{p}) \pmod{p}$.

Lemma 2.2, with $m = 2k$, transforms the second sum modulo p^2 into

$$p\sum_{k=0}^{h} \frac{(20k+7)(-4)^k b_k^2}{(4k+1)b_{2k}} = pR_p \equiv 7p \pmod{p^2}$$

by Theorem 2.4. Combining the two contributions proves

$$S(p) \equiv p\left(7 + 10\left(\frac{-1}{p}\right)\right) \pmod{p^2}.$$

□

3 Consequences, scope, and verification

Corollary 3.1. *For an odd prime p,*

$$p^2 \mid S(p) \quad \Longleftrightarrow \quad p \in \{3, 17\}.$$

For every odd prime $p \notin \{3, 17\}$, one has $v_p(S(p)) = 1$.

Proof. If $p \equiv 1 \pmod 4$, Theorem 2.1 gives $S(p) \equiv 17p \pmod{p^2}$; this is zero precisely when $p = 17$. If $p \equiv 3 \pmod 4$, it gives $S(p) \equiv -3p \pmod{p^2}$; this is zero precisely when $p = 3$. For every other odd prime the residue has valuation exactly one. □

Together with the separate all-integer divisibility argument in [6], the present theorem supplies proofs of both assertions in Conjecture 5.8(ii). This statement concerns the mathematical content of the two research manuscripts. No result about the three unweighted cases in Conjecture 5.8(i) is claimed here, and no assertion modulo p^3 is inferred.

The essential denominator issue is explicit in Theorem 2.4: some individual summands have valuation -1, so termwise reduction of the original reciprocal sum modulo p would be invalid. The exact finite identity (3) first replaces that sum by terms in $\mathbb{Z}_{(p)}$. The valuation estimates and the endpoint values then give its residue. This is the arithmetic step beyond a change of notation or a formal identity alone.

A bounded search of the exact sequence, the unchanged weight, equivalent binomial notation, and the reciprocal sum did not identify a previous complete proof of this assertion. That search does not certify historical firstness. The WZ technique and the motivating kernels are established mathematics and are credited above; we do not claim to have invented that general method. This is a newly prepared proof manuscript, checked internally by independent readings and exact arithmetic. It has not undergone external peer review or proof-assistant verification.

The program in the appendix audits the rational certificate, the finite telescoping identity, all boundary valuations, the universal inner-sum congruence, and the original weighted congruence. The reported run passed 81 exact certificate checks and all stated tests at the 25 odd primes at most 101. These finite computations are not assumptions of any theorem above.

A Exact reproducibility checks

The following complete Python program uses only the standard library. It is included inside this single TEX file. It checks the proof's transcription using exact integers and rational numbers; the unbounded result is supplied by the algebraic proof.

```
"""Exact checks for the all-odd-prime companion congruence.
The proof is symbolic; these checks audit its transcription only.
"""
from fractions import Fraction as Q
from math import comb, prod, isqrt


def poch(a, k):
    return prod((a+j for j in range(k)), start=Q(1))


def b(n):
    return comb(2*n, n)


def c(n):
    return Q((-4)**n*b(n)**2, (4*n+1)*b(2*n))


def H(n, k):
    return c(n)*poch(Q(-4*n-1, 2), k)*poch(Q(2*n+1, 2), k)/poch(Q(1, 2), k)**2


def D(n, k):
    return (20*n+2*k+7)*H(n, k)


def E(n, k):
    if n == -1:
        return Q(0)
    return Q(16*(2*n+1)**2, 4*n-2*k+3)*H(n, k)


def vp(x, p):
    x = Q(x)
    if not x:
        return float('inf')
    ans = 0
    n, d = abs(x.numerator), x.denominator
    while n % p == 0:
        n //= p
        ans += 1
    while d % p == 0:
        d //= p
        ans -= 1
    return ans


for n in range(9):
    for k in range(1, 10):
        assert D(n, k)-D(n, k-1) == E(n, k)-E(n-1, k)

primes = [p for p in range(3, 102, 2)
          if all(p % q for q in range(2, isqrt(p)+1))]
partial = [0]
```

```
for n in range(101):
    a = sum((-4)**k*comb(n, k)**2*comb(n-k, k)**2
            for k in range(n//2+1))
    partial.append(partial[-1]+(20*n+17)*a)

for p in primes:
    h = (p-1)//2
    R = sum((20*n+7)*c(n) for n in range(h+1))
    assert R == sum(D(n, h) for n in range(h+1))-sum(E(h, k) for k in range(1, h+1))
    assert vp(R-7, p) >= 1
    assert D(0, h) == Q(-(p+6), p-2)
    assert H(h, h) == 4**h*b(h)*prod((Q(p+2*i, 2*i+1) for i in range(1, h)), start=Q(1))
    assert vp(H(h, h)+1, p) >= 1
    assert all(vp(H(n, h), p) == 1 for n in range(1, h))
    assert all(vp(E(h, k), p) == 2 for k in range(1, h+1))
    assert partial[p] % (p*p) == p*(7+10*(-1)**h) % (p*p)
    for m in range(p):
        T = sum(comb(n, m)**2 for n in range(m, p))
        assert vp(T-Q(p, (2*m+1)*b(m)), p) >= 2

print('PASS: 81 exact WZ identities, including n=0 boundary')
print('PASS: reciprocal congruence and all endpoint valuations at', len(primes), 'odd primes <=
    101')
print('PASS: original companion congruence at the same primes')
print('PASS: every inner-sum congruence for m < p at those primes')
```

References

[1] Zhi-Wei Sun, *On Apéry numbers and generalized central trinomial coefficients*, arXiv:1006.2776v11, 19 August 2010, Conjecture 5.8(ii), printed p. 33. https://arxiv.org/abs/1006.2776v11. The specified version contains the conjecture; the subsequent withdrawal notice concerns reorganization into other manuscripts.

[2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A179524, contributed by Zhi-Wei Sun, 17 July 2010; accessed 8 November 2026. https://oeis.org/A179524.

[3] Jesús Guillera, *Hypergeometric identities for 10 extended Ramanujan-type series*, Ramanujan Journal **15** (2008), 219–234. https://doi.org/10.1007/s11139-007-9074-0. Preprint: https://arxiv.org/abs/1104.0396.

[4] Zhi-Wei Sun, *A refinement of a congruence result by van Hamme and Mortenson*, Illinois Journal of Mathematics **56** (2012), no. 3, 967–979. https://arxiv.org/abs/1011.1902. See Section 3 for the motivating WZ pair.

[5] Qinghu Hou, Haihong He, and Xiaoxia Wang, *Ramanujan-inspired series for $1/\pi$ involving harmonic numbers*, Journal of Difference Equations and Applications **30** (2024), 775–786. https://doi.org/10.1080/10236198.2024.2325492. Preprint: https://arxiv.org/abs/2305.00498v4, equation (4.1).

[6] Research manuscript prepared with ChatGPT for Keio, *A Proof of Sun's Divisibility Conjecture for A179524*, unpublished research manuscript, 8 November 2026, 13 pp. The present paper does not use its theorem as an input.